

Adaptive Sensitivity Configuration for Remote Examination Monitoring Systems Driven by Examination Scheme Metadata and Candidate Accommodation Declarations

Author: Chintan Dave

Affiliation: Independent researcher, Ahmedabad, India

Contact: ce.chintan@gmail.com

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Abstract

This disclosure describes a method for dynamically configuring the sensitivity of behavioral monitoring systems used during remotely-administered examinations. Conventional remote-proctoring systems apply a single, fixed configuration of flag thresholds, event weights, and suppression rules to every examination they monitor, irrespective of the nature of the examination or the declared accommodations of the candidate. This produces well-documented harms: open-book examinations are penalized for tab-switching behaviors that are explicitly permitted by examination rules; candidates with declared disabilities or assistive-technology requirements are penalized for behaviors that constitute their accommodations; and high-stakes licensure examinations are subjected to the same coarse tolerance as low-stakes practice quizzes. This disclosure describes an architecture in which a per-session sensitivity configuration is computed at examination-start time from (i) examination-scheme metadata describing the rules and content type of the examination and (ii) a candidate-specific accommodation declaration, and is applied to a real-time monitoring pipeline such that events of suppressed classes are logged for audit purposes but excluded from risk-score generation.

1. Background

Remote examination monitoring systems are typically configured with a fixed set of detector rules, event weights, and flag thresholds. The configuration applies uniformly to every examination administered through the system. Some commercial systems permit examination administrators to manually adjust a small number of overall

sensitivity settings, but these adjustments are static, coarse-grained, and applied at the level of an examination program rather than at the level of an individual candidate session.

This fixed-configuration approach produces several well-documented operational problems. Open-book examinations, in which candidates are explicitly permitted to consult notes or external materials, are nonetheless monitored under rules that flag tab-switching, application-window changes, and gaze excursion away from the screen. Examinations with structured oral components, such as language-proficiency speaking sections, are monitored under rules that flag candidate speech. Performance-based examinations involving manipulation of physical or digital tools are monitored under rules calibrated for sit-still-and-answer behavior. Examinations administered to candidates with declared accommodations — such as candidates with attention-deficit conditions who are permitted scheduled movement breaks, candidates with visual impairments using screen-reader software, candidates with motor impairments using alternative input devices, or candidates with anxiety conditions accompanied by approved support animals — are monitored under rules that flag the very behaviors the accommodations are designed to permit.

The cumulative effect of fixed-configuration monitoring is a baseline false-positive rate that disproportionately affects candidates whose lawful examination behavior diverges from a narrow notional default. This is documented in the academic literature on remote-proctoring fairness and is the subject of binding recommendations from European data-protection authorities.

The technique disclosed herein addresses this problem by computing a per-session monitoring configuration from structured metadata describing the examination and the candidate's declared accommodations.

2. Technique

2.1 Inputs

At the commencement of an examination session, the monitoring system receives three inputs that together determine the session's monitoring configuration:

(a) Examination scheme metadata. A structured record describing the examination, containing at least: - an open-book / closed-book attribute; - a permitted-materials set (e.g., calculator, scratch paper, named reference works); - a permitted-input-method set (e.g., keyboard, mouse, touch-screen, voice dictation); - a permitted-behavior set (e.g., speech permitted for oral component, standing permitted, scheduled breaks permitted); - an examination-content lexicon (a list of keywords whose detection in audio would constitute a substantive integrity concern); - a stakes designation (e.g., low-stakes practice, mid-stakes course assessment, high-stakes credential, regulated licensure); - an applicable accreditation standard (e.g., ISO/IEC 17024, NCCA, jurisdiction-specific equivalents).

(b) Candidate accommodation declaration. A structured record listing the candidate's pre-approved accommodations, each mapping a declared condition to one or more suppressible monitoring-event classes. Example mappings include:

Declared accommodation	Suppressed monitoring-event classes
Scheduled movement break (e.g., for ADHD)	Standing-up event; face-out-of-frame event up to N seconds; repeated-posture-change event
Screen-reader software (e.g., for visual impairment)	Unrecognized-process event for declared screen-reader process; high-screen-reading-frequency event
Alternative input device (e.g., for motor impairment)	Unrecognized-input-device event; atypical-keystroke-cadence event
Approved support animal	Animal-vocalization event; animal-in-frame event
Reader/scribe (e.g., for dyslexia)	Additional-voice event when matching declared reader's voice profile
Approved bathroom breaks	Face-out-of-frame event up to declared duration

The accommodation declaration is established prior to the examination via the candidate's pre-examination administrative workflow, and accompanying documentation is held by the certification body in accordance with applicable disability-accommodation procedures.

(c) A default sensitivity profile. A baseline configuration of event weights, flag thresholds, and suppression rules used in the absence of overriding metadata.

2.2 Configuration computation

A per-session sensitivity configuration C is computed as a function:

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 $C = f(\text{examination\_metadata}, \text{accommodation\_declaration}, \text{default\_profile})$ 
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The function f applies the following operations:

1. Start with the default profile.
2. For each examination-scheme attribute, apply the corresponding override to the default profile. For example, if the examination is open-book, set the tab-switch event weight to zero and clear the application-allowlist enforcement.
3. For each accommodation in the accommodation declaration, apply the corresponding suppression rule. For example, if the candidate has a declared scheduled-movement-break accommodation, configure the face-out-of-frame detector to log but not weight events of duration shorter than the declared break duration.
4. For the stakes designation, apply a global sensitivity multiplier — higher stakes apply a tighter overall threshold for high-tier flagging.
5. For the applicable accreditation standard, apply standard-specific requirements such as mandatory human review for any adverse action, mandatory two-reviewer concurrence, mandatory disclosure of AI use to the candidate, and standard-specific record-retention rules.

The resulting configuration C is a complete, self-contained specification of all monitoring rules to be applied to this specific session for this specific candidate.

2.3 Application to real-time pipeline

The configuration C is loaded into the real-time monitoring pipeline at session start. During the session, detector modules emit raw events as usual. A configuration-aware filter examines each event against C:

- If the event's class is in the suppressed set of C, the event is written to the session log with a marker indicating it was suppressed, and is excluded from risk-score computation.
- If the event's class has a configuration-specified weight or threshold in C, the configuration-specified value is used in place of the default.
- All other events flow through the pipeline as usual.

2.4 Audit logging

The complete configuration C is written to an immutable audit log alongside the session risk score and any flagged events. This is critical for subsequent appeal review: the audit log must demonstrate not only what behavior was observed but also what monitoring configuration was applied at the time, including which suppression rules were in effect and on what authority (the accommodation declaration document). Without this audit record, a candidate appealing an adverse finding cannot verify that the system correctly honored their accommodations.

2.5 Configuration versioning

The default sensitivity profile and the function f are versioned. The version identifier is included in the audit log so that any subsequent review can reconstruct exactly what monitoring logic was applied to the session, even if the system has been updated since.

3. Variant embodiments

The following variants are within the scope of the disclosed technique and are explicitly published as prior art:

- (a) Application of the technique to live-proctored as well as record-and-review remote examinations.
- (b) Application to in-person computerized examinations administered in test centers with networked monitoring.
- (c) Implementation of the function f as a rules engine, as a configuration template-and-merge operation, as a machine-learning model trained on adjudicated session outcomes, or as a hybrid of these approaches.
- (d) Extension of the accommodation declaration to include candidate self-attestation at the time of session start, such as "I have a service dog present in the room today," which the system records and applies as a session-specific override after appropriate verification or as a candidate-attested suppression with appropriate audit trail.
- (e) Federation across multiple certification bodies sharing structured accommodation schemas and examination-scheme metadata schemas.
- (f) Integration with disability-accommodation administrative systems that maintain candidate accommodation records.
- (g) Use of the configuration C as input to a fusion engine that performs multi-signal correlation, including correlation between behavioral telemetry and psychometric anomaly signals.
- (h) Adjustment of configuration C during the session in response to declared mid-session events such as a candidate-initiated technical-difficulty escalation, with appropriate audit logging.

- (i) Application of the technique to examinations administered in multiple languages, where the examination-content lexicon and the candidate-speech detection are language-specific.
 - (j) Privacy-preserving variant in which the accommodation declaration is held in encrypted form and decrypted only at session start by an authorized subsystem.
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4. Operational considerations

4.1 Accommodation declaration integrity

The accommodation declaration must be established through an administrative process that includes appropriate documentation review, in accordance with applicable disability law (e.g., the Americans with Disabilities Act, the Rights of Persons with Disabilities Act 2016 in India, the Equality Act 2010 in the United Kingdom, and equivalent statutes). The monitoring system does not independently verify the merits of declared accommodations; it applies the suppressions specified in the accommodation declaration as established by the certification body's administrative process.

4.2 Default-profile calibration

The default sensitivity profile should be calibrated periodically based on confirmed false-positive rates observed in human-review adjudications. The calibration process should be stratified by demographic categories where consent and statistical power permit, to ensure the default profile does not embed demographic disparities.

4.3 Examination-scheme metadata governance

The examination-scheme metadata must be defined and maintained by the certification body's examination committee or psychometric staff, in accordance with the certification body's documented procedures. Changes to the metadata should be versioned and audit-logged. Any change to monitoring sensitivity that would affect candidates already enrolled in an active examination cycle should be reviewed for fairness implications.

4.4 Reviewer training

Human reviewers who adjudicate flagged sessions must be trained on the configuration model, including how to read the configuration audit record, how to verify that suppressed events were correctly suppressed, and how to escalate cases where the configuration appears to have been mis-applied.

5. Technical effects

The disclosed technique provides the following measurable improvements over fixed-configuration remote-proctoring systems known in the art:

- Reduction in the false-positive rate for examinations whose rules differ from a narrow notional default (open-book, oral, performance-based, etc.).
- Reduction in the false-positive rate for candidates with documented accommodations.
- Reduction in human-reviewer workload by suppressing flags whose triggering events represent permitted or accommodated behavior.

- Improved auditability of monitoring decisions, by maintaining a complete record of which configuration was applied to each session.
 - Improved alignment with ISO/IEC 17024:2026 fairness and impartiality requirements, EU AI Act human-oversight requirements, and applicable disability-accommodation statutes.
 - Improved candidate trust, by making the monitoring system's treatment of the candidate's specific situation transparent and verifiable.
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